

Functional genetic report · longevity

Detoxification Phase I and II

Sérgio F · Report date: 02-10-2026

How the body activates and eliminates toxic substances (smoking, burned meat, solvents, pesticides, plastics), estrogens and drugs. Only genes (chip + local imputation + bilirubin score).

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Summary

39

SNPs analyzed

31

in the chip

6

imputed ($DR2 \geq 0.8$)

2

SNP not found in the data

1

Polygenic scores

Detoxification has two phases. In phase I, the cytochromes P450 oxidize the molecule, which sometimes makes it more reactive (e.g., aromatic hydrocarbons → epoxides that damage DNA). In the II phase conjugation enzymes (GST, NAT2, UGT, SULT) and NQO1 neutralize these intermediates and make them soluble to be eliminated. The balance between the two phases, and the ability to produce glutathione (via NRF2), determines how much oxidative damage and DNA is behind.

In synthesis: the phase I of toxic activation is favourable (CYP1A1 and CYP1B1 without the activating variants; NQO1 and NRF2 normal). The weak points are in the II phase and in the lipid protection: slow acetylator NAT2 (*5/*5), GSTA1*B/*B (low expression), a low glutathione synthesis allele (GCLM) and a low activity PON1 profile. In the drugs, CYP2C9*3 and CYP3A5 expressor (relevant to rapamycin) stand out.

Favourable

- **Low toxic activation:** CYP1A1 Ile462Val and MspI absent; CYP1B1 Leu432/Leu432 (less 4-hydroxiestradiol).
- **NQO1 Pro187/Pro187** (rs1800566 GG): normal activity (quinones, regeneration of CoQ10 and vitamin E).
- **NRF2, GCLC and catalase** without the low expression alleles.
- **Favourable Pharmacogenomics:** Probable SLCO1B1 *1/*14 (augmented function without rs4149056 myopathy allele): statins without genetic restriction; CYP2C19 *1/*1, without CYP2D6*4 or CYP3A4*22.
- **UGT1A1 *1/*28 (Gilbert light)** with polygenic bilirubin in the 71st percentile: more antioxidant bilirubin.

Unfavourable

- **NAT2 slow acetylator** (*5/*5; rs1495741 AA): eliminates the aromatic and heterocyclic amines (grid/burned meat, smoke). Common but unfavourable phenotype.
- **GSTA1 *B/*B** (rs3957357 AA): lower GST alpha in the liver (lipid peroxidation aldehydes).
- **GCLM -588 heterozygote** (rs41303970 GA): a low synthesis allele of glutathione.
- **PON1 of low activity:** Q192/Q192, M55/M55 and -108T/T (less paraoxonase in HDL; more sensitive to organophosphates).
- **CYP2C9 *1/*3** (Lower NSAIDs and coumarins) and **CYP3A5 *1/*3 expressor** (lower exposure to rapamycin per mg). GSTM1/GSTT1 (deletions) not visible.

Interpretation: The profile is not of easy "intoxication": the activating phase I is calm and NQO1/NRF2 are intact. The weakness is in the conjugation (slow NAT2, low GSTA1, a little limited glutathione) and in PON1. The strategy is to reduce exposure to amines from burned meat, tobacco and pesticides, and to support glutathione and NRF2 (crucifers).

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 31 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

There is only one polygenic score useful in this topic: total bilirubin (PGS000697, 1 159 variants). A genetically higher bilirubin is antioxidant and is associated with less cardiovascular disease, so here a high percentile is favourable.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Total bilirubin (1 159 variants)	PGS000697	1 061 de 1 159 (92%)	p71	Typical / no judgement

Phase I: Toxic activation cytochromes P450 (CYP1A1, CYP1A2, CYP1B1, CYP2E1) and AhR

CYP1A1 and CYP1B1 activate polycyclic aromatic hydrocarbons (smoker, grilled) and, in the case of CYP1B1, transform estradiol into 4-hidroxiestradiol. CYP1A2 metabolizes caffeine, melatonin and heterocyclic amines. AhR is the sensor that induces these enzymes.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CYP1A1 Ile462Val (*2C)	rs1048943	TT	chip	C	0	0.03	Val462: CYP1A1 more active; more activation of aromatic hydrocarbons (smoker, burned meat) in metabolites that damage DNA. (moderate evidence)	Typical / no judgement
CYP1A1 MspI (*2A)	rs4646903	AA	imputation DR2=0.99	G	0	0.10	CYP1A1*2A: higher inductibility of CYP1A1. (weak evidence)	Typical / no judgement
CYP1A2 *1F (-163C>A)	rs762551	CA	chip	C	1	0.32	CYP1A2: A allele is very inducible (fast caffeine and melatonin metabolizer); C is slower. Slower : caffeine and melatonin last longer, but less activation of heterocyclic amines. (moderate evidence)	Typical / no judgement
CYP1B1 Leu432Val	rs1056836	GG	chip	C	0	0.42	CYP1B1 Val432: plus 4-hydroxylation of estradiol (catecolestrogen more reactive) and activation of aromatic hydrocarbons. (weak evidence)	Favourable
CYP2E1 *5B (RsaI)	rs2031920	CC	chip	T	0	0.04	CYP2E1 c2: altered transcription (solvents, acetaminophen, nitrosamines). (weak evidence)	Typical / no judgement
CYP2E1 *6 (DraI)	rs6413432	TT	imputation DR2=1.00	A	0	0.10	CYP2E1 DraI: altered activity. (weak evidence)	Typical / no judgement
AHR	rs4410790	CC	chip	C	2	0.61	Aryl-hydrocarbon receptor (dioxin sensor, broccoli indoles and tryptophan): C allele is associated with more induction of CYP1A1/1A2 and higher consumption of coffee (faster caffeine metabolism). (moderate evidence)	Typical / no judgement

Favourable

- CYP1A1 rs1048943 TT and rs4646903 AA (without *2C or *2A)
- CYP1B1 rs1056836 GG (Leu/Leu)
- CYP2E1 without *5B or *6

Unfavourable

- None relevant

Interpretation: Phase I of calm activation. CYP1A2 *1F heterozygous (AC): intermediate metabolizer of caffeine and melatonin (melatonin lasts a little longer). AhR rs4410790 CC: allele of greater inductibility (faster response to grills, smoke and indoles of the crucifers).

Phase I: drug enzymes (CYP3A, CYP2C, CYP2D6) and transporters

The enzymes that eliminate the drugs. The current drugs most dependent on them: rapamycin (CYP3A4/3A5, P-glycoprotein), tadalafil (CYP3A4), isotretinoin (CYP2C8/3A4) and melatonin (CYP1A2).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CYP3A5 *3	rs776746	CT	chip	C	1	0.94	CYP3A5*3/*3 = non-expressor (the most common in the reference population). Metabolizes rapamycin, tadalafil and isotretinoin together with CYP3A4. (strong evidence)	Typical / no judgement
CYP3A4 *22	rs35599367	GG	chip	A	0	0.05	CYP3A4*22: ~50% lower expression; higher levels of rapamycin/tacrolimus. (strong evidence)	Favourable
CYP2C9 *2	rs1799853	CC	chip	T	0	0.13	CYP2C9*2: reduced activity (NSAIDs, warfarin, sulfonylureas). (strong evidence)	Favourable
CYP2C9 *3	rs1057910	AC	chip	C	1	0.07	CYP2C9*3: very reduced activity (NSAIDs such as ibuprofen and celecoxib last longer; risk of gastrointestinal and renal effects). (strong evidence)	Unfavourable
CYP2C19 *2	rs4244285	GG	chip	A	0	0.14	CYP2C19*2: loss of function (clopidogrel, PPI, some antidepressants). (strong evidence)	Favourable
CYP2C19 *17	rs12248560	CC	chip	T	0	0.22	CYP2C19*17: increased activity. (strong evidence)	Typical / no judgement
CYP2D6 *4	rs3892097	CC	chip	T	0	0.19	CYP2D6*4: allele without function (codein, tramadol, many antidepressants and betablockers). (strong evidence)	Favourable
CYP2C8 *3 (R139K)	rs11572080	CC	imputation DR2=1.00	T	0	0.12	CYP2C8*3: altered activity (isotretinoin/retinoic acid, pioglitazone). (moderate evidence)	Typical / no judgement
ABCB1 C3435T	rs1045642	AA	chip	A	2	0.52	P-glycoprotein (rapamycin efflux pump, tadalafil and loperamide): small and inconsistent effect. (weak evidence)	Typical / no judgement
SLCO1B1 *5 (V174A)	rs4149056	TT	chip	C	0	0.15	SLCO1B1 V174A: lower hepatic uptake of statins (sinvastatin) → more myopathy. (strong evidence)	Favourable

Favourable

- Probable SLCO1B1 *1/*14 (augmented function without rs4149056 myopathy allele): statins without genetic restriction
- CYP2C19 *1/*1, CYP2D6 without *4 (probable profile: *1/*2), CYP3A4 without *22, CYP2C9 without *2

Unfavourable

- CYP2C9 *1/*3 (rs1057910 AC): Long term NSAIDs and slower coumarins
- CYP3A5 *1/*3 (rs776746 CT): expression, ~10% of the reference population; lower exposure to rapamycin per mg

Interpretation: ABCB1 3435 TT (rs1045642 AA): P-glycoprotein possibly a little less expressed, small effect. CYP2C8*3 absent (isotretinoin without alteration). Together: CYP3A5 expresser and P-gp TT have opposite effects on rapamycin; only one blood level says the real effect.

Phase II: glutathione S-transferases

Glutathione S-transferases bind glutathione to electrophiles (epoxides, aldehydes, quinones, metals). GSTM1 and GSTT1 are often deleted, but the chip does not read deletions.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
GSTP1 Ile105Val	rs1695	AG	chip	G	1	0.34	GSTP1 Val105: lower conjugation of some electrophiles (benzopyrene-diol-epoxide), higher than others. (weak evidence)	Typical / no judgement
GSTP1 Ala114Val	rs1138272	CC	chip	T	0	0.07	GSTP1 Val114: reduced activity. (weak evidence)	Typical / no judgement
GSTA1 *B (-69C>T)	rs3957357	AA	chip	A	2	0.43	GSTA1*B: lower hepatic expression of GST alpha (conjugation of lipid peroxidation aldehydes, such as 4-HNE). (weak evidence)	Unfavourable
GSTM1 (marker)	rs366631	AA	chip	—	—	—	SNP of the GSTM1/GSTM2 region: does not serve to infer the null GSTM1 deletion, which the chip does not read. (weak evidence)	Typical / no judgement

Favourable

- GSTP1 Ala114/Ala114 (rs1138272 CC)

Unfavourable

- GSTA1 *B/*B (rs3957357 AA): low expression
- GSTP1 Ile105Val heterozygous (typical)

Interpretation: GSTA1 is the most abundant GST in the liver and treats 4-hydroxynonenal (Omega-6/ Omega-3 peroxidation aldehyde). With GSTA1*B/*B, it is appropriate to protect fatty acids from oxidation (fresh and refrigerated fish oil without reheated fried). GSTM1/GSTT1: unknown.

Phase II: acetylation (NAT2), quinones (NQO1), sulfate, glucuronidation and epoxides

Acetylation (NAT2), reduction of quinones (NQO1), sulfate (SULT1A1), glucuronidation (UGT1A1) and hydrolysis of epoxides (EPHX1).

Polygenic scores of this theme: Total bilirubin (1 159 variants) (PGS000697): the 71st percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
NAT2 *5 (341T>C)	rs1801280	CC	chip	C	2	0.45	NAT2*5: slow allele.	Typical / no judgement
NAT2 *6 (590G>A)	rs1799930	GG	chip	A	0	0.28	NAT2*6: slow allele.	Typical / no judgement
NAT2 *7 (857G>A)	rs1799931	GG	chip	A	0	0.02	NAT2*7: slow allele.	Typical / no judgement
NAT2 phenotype marker	rs1495741	AA	chip	A	2	0.75	Best single SNP of the NAT2 phenotype: AA : slow acetylator (aromatic and heterocyclic amines of grilled meat and tobacco, isoniazid, hydralazine, sulfonamides). (strong evidence)	Typical / no judgement
NQO1 P187S (*2)	rs1800566	GG	chip	A	0	0.20	NQO1*2/*2 : no activity (reduction of quinones, regeneration of CoQ10 and vitamin E, protection against benzene). (strong evidence)	Favourable
SULT1A1 Arg213His (*2)	rs9282861		SNP not found in the data	—	—	—	SULT1A1*2: lower sulfate (phenols, catechelestrogens); the gene also has a variation in the number of copies, which the chip does not read. (weak evidence)	SNP not found in the data
UGT1A1 *80 (*28 marker)	rs887829	CT	chip	T	1	0.30	UGT1A1*80/*28 (Gilbert): lower glucuronidation → higher bilirubin (antioxidant, associated with less cardiovascular disease); some slower drugs (irinotecan). (strong evidence)	Typical / no judgement
EPHX1 Tyr113His	rs1051740	TT	chip	C	0	0.30	Epoxide hydrolase His113: ~40% lower activity (epoxy aromatic hydrocarbons). (weak evidence)	Typical / no judgement
EPHX1 His139Arg	rs2234922	AG	chip	G	1	0.17	Epoxide hydrolase Arg139: ~25% higher activity. (weak evidence)	Typical / no judgement

Favourable

- NQO1 rs1800566 GG (normal activity)
- EPHX1 Tyr113/Tyr113 and 1 allele Arg139 fast (normal to high activity)
- UGT1A1 *1/*28: higher bilirubin (antioxidant); bilirubin p71 score

Unfavourable

- NAT2 *5/*5: slow acetylator (rs1801280 CC, rs1495741 AA)

Interpretation: Slow NAT2 is the most relevant finding of this phase: heterocyclic amines of cooked meat and aromatic amines of tobacco are more available for activation. SULT1A1*2 not found in the data (the gene has copy variation).

NRF2, glutathione and antioxidant enzymes (catalase, SOD2, MPO, PON1)

NRF2 binds the genes of the II phase and glutathione. GCLC and GCLM produce glutathione; catalase and SOD2 treat peroxides; MPO produces oxidants in neutrophils; PON1 protects HDL and organophosphate hydrolyses.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
NFE2L2 (NRF2) -617C>A	rs6721961	GG	imputation DR2=1.00	T	0	0.12	Promoter of NRF2 (master regulator of detoxification and glutathione): Allele A reduces the expression of NRF2. (moderate evidence)	Typical / no judgement
GCLM -588C>T	rs41303970	GA	chip	A	1	0.15	Glutamate-cysteine ligase (glutathione synthesis) modulating subunit: T reduces expression ~50%. (moderate evidence)	Unfavourable
GCLC -129C>T	rs17883901	GG	chip	A	0	0.08	Catalytic subunit of glutathione synthesis: T reduces oxidative stress response. (moderate evidence)	Typical / no judgement
CAT -262C>T	rs1001179	CC	chip	T	0	0.23	Catalase: the T allele is associated with a different activity from catalase (hydrogen peroxide). (weak evidence)	Typical / no judgement
SOD2 Val16Ala	rs4880	GG	chip	—	—	—	Mitochondrial SOD2: Ala16 is better in mitochondria (plus H2O2, plus catalase/GPX charge); Less efficient Val16. Effect depends on balance. (weak evidence)	Typical / no judgement
MPO -463G>A	rs2333227		SNP not found in the data	T	—	—	Myeloperoxidase: allele A (-463) reduces expression (less hypochlorous acid and activation of pro-carcinogens). (weak evidence)	SNP not found in the data
PON1 Q192R	rs662	TT	chip	C	0	0.30	Paraoxonase 1 (HDL; organophosphate hydrolyses and oxidized lipids): R192 better hydrolyzes the paraoxon, Q192 better diazoxon and sarin. (moderate evidence)	Typical / no judgement
PON1 L55M	rs854560	TT	imputation DR2=1.00	T	2	0.36	PON1 M55: lower concentration of PON1. (weak evidence)	Typical / no judgement
PON1 -108C>T	rs705379	AA	imputation DR2=0.97	A	2	0.50	Promoter PON1: T(-108) reduces expression ~2 times (less PON1 in HDL). (moderate evidence)	Unfavourable

Favourable

- NFE2L2 -617 GG (normal NRF2 expression)
- GCLC -129 GG and catalase -262 CC

Unfavourable

- GCLM -588 GA (1 low glutathione synthesis)
- PON1 Q192/Q192, M55/M55 and -108T/T: the lowest activity profile and concentration of PON1

Interpretation: SOD2 Ala16/Ala16 (rs4880 GG): more superoxide converted into H2O2 within the mitochondria, so catalase and normal GPX are important. MPO -463 not found in the data. The intact NRF2 means that Sérgio F responds well to food inducers (sulforaphan).

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
DNA Damage and DNA Repair Deficit / Somatic Mutation Accumulation	Slow NAT2; unknown GSTM1/GSTT1	CYP1A1 and CYP1B1 without activating variants; Normal NQO1; Normal EPHX1	Balanced: less activation, slower conjugation of amines.
Oxidative stress / Redox imbalance	GSTA1*B/*B; GCLM -588 het; PON1 low	NRF2, GCLC, normal catalase and NQO1; higher bilirubin (UGT1A1*28)	Slightly unfavourable, compensable.
Vascular Aging	PON1 of low activity	Higher bilirubin (light Gilbert)	Neutral.
Decline of the Estrogen Signal / Endocrine Deregulation	—	CYP1B1 Leu/Leu (less 4-hydroxiestradiol)	Favourable.
Chronic Inflammation	Very inducible AhR (CC)	Normal NRF2	Neutral.
Oxidative Phosphorylation Decline	SOD2 Ala/Ala (more H2O2 mitochondrial)	Normal NQO1 (regenerates CoQ10)	Neutral.
Peroxisomal dysfunction	—	Catalase -262 CC	Favourable.
Sleep disturbance	Intermediate CYP1A2 (caffeine lasts longer)	—	Mild weakness.
Metabolic Homeostasis decline (pharmacogenomic)	CYP2C9*3; CYP3A5 expressor	Likely SLCO1B1 *1/*14 (augmented function); CYP2C19 and CYP2D6 normal	It matters to NSAIDs and rapamycin.
Sarcopenia	—	Probable SLCO1B1 *1/*14 (augmented function without rs4149056 myopathy allele)	Favourable if you need statin to lower LDL/ApoB.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Crucifers daily	Broccoli, broccoli shoots, kale shoots, arugula, cress: induce NRF2, GST and NQO1. More fiber, always (biliary and toxic acids eliminated by bile). <i>suggestion (not applied)</i>	Intact NRF2; Low GSTA1; Slow NAT2
Sulforaphane	Optional: broccoli shoot extract with myrosinase (glucorphanin 30-60 mg/day) on days without crucifers. <i>suggestion (not applied)</i>	Normal NRF2; GCLM het
Glutathione	Glycine 3-6 g/day ± NAC 600 mg (GlyNAC) and enough protein with cysteine. <i>suggestion (not applied)</i>	GCLM -588 het; GSTA1*B/*B
Meat and cooking	Avoid burnt meat, very toasted and smoked grills; cook at the lowest temperature, marinate, join herbs (alecrim) and crucifers on the same dish. <i>suggestion (not applied)</i>	NAT2 slow + AhR inducible
Exposures	Zero tobacco (including passive); washing well the fruit and vegetables and prefer biological in the most contaminated by pesticides; not heating food in plastic; ventilating when using solvents. <i>suggestion (not applied)</i>	Low PON1; slow NAT2; GST
Protecting lipids	Fresh, refrigerated fish oil without rancid smell; extra virgin olive oil and pomegranate oil (suffer PON1 activity); without reheated fried oil. <i>suggestion (not applied)</i>	GSTA1*B/*B; PON1 low
Sauna	3-4 sessions/week: increases the elimination of some toxics (BPA, phthalates, metals) by sweat and induces heat shock proteins. <i>suggestion (not applied)</i>	Slower II phase
Rapamycin	A serum level of sirolimus, once, to calibrate the dose (CYP3A5 expressor). Avoid grapefruit and hypericon, or always use them in the same way, because they greatly change the levels of rapamycin and tadalafil. <i>suggestion (not applied)</i>	CYP3A5 *1/*3; normal CYP3A4
Analyses	Avoid long-term NSAIDs (pyroxicam, meloxicam, celecoxib) and chronic use of ibuprofen; if necessary, paracetamol in low dose and for a short time (ALT is high). <i>suggestion (not applied)</i>	CYP2C9 *1/*3
Caffeine and melatonin	Cafein only in the morning; melatonin continues as it is (the intermediate CYP1A2 makes it last a little longer). Avoid fluvoxamine and ciprofloxacin (very good melatonin). <i>suggestion (not applied)</i>	CYP1A2 *1F heterozygous
Statin, if necessary	The lowest LDL/ApoB is always better; the likely SLCO1B1 *1/*14 (augmented function, without the rs4149056 myopathy allele) does not put genetic restrictions on a statin (with CYP2C9*3, prefer rosuvastatin or atorvastatin to fluvastatin). <i>suggestion (not applied)</i>	Probable SLCO1B1 *1/*14; CYP2C9*3
Coordination with the plan	Curcumin, luteolin, apigenin and quercetin in pause while ALT is high; sinusolytic pulse only physetin + quercetin with normal ALT. D3 10 000 IU and omega-3 3.2-4 g/day as in the plane. No telomerase activators. <i>suggestion (not applied)</i>	V2 plane rules

Limitations

- GSTM1 and GSTT1 deletions and copy variations of CYP2D6, SULT1A1 and UGT2B17 are not read by the chip; the profile does not know them.
- Haplotypes (NAT2, CYP2D6, PON1) were inferred from isolated SNPs; the phase of alleles is not directly observed.
- The effect of these variants is modest and depends heavily on exposure (diet, smoking, profession, drugs). Without exposure, a slow genotype may not have consequence.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use blood analysis or functional tests.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Frequency of alleles calculated locally in the reference population (n=633); the effect allele of each SNP was confirmed by the expected frequency.
- Alleles of effect cured from literature and pharmacogenomic tables (nomenclature of star alleles).
- Score: PGS000697 (total bilirubin, Sinnott-Armstrong 2021). Percentage vs. reference population (n=633) with exactly the same variants.
- Main references: Hein 2000 (NAT2); García-Closas 2005 (NAT2 and bladder, Lancet); Nebert and Dalton 2006 (CYP1, Nat Rev Cancer); Ross 2000 (NQO1); 2001 (GSTA1*B) collars; Nakamura 2002 (GCLM -588); Marzec 2007 (NRF2 -617); Mackness 2004 and Deakin 2004 (PON1); Bosma 1995 (UGT1A1*28); Kuehl 2001 (CYP3A5*3, Nat Genet); Theken 2020 (CPIC CYP2C9 and NSAIDs); Cooper-DeHoff 2022 (CPIC SLCO1B1 and statins); Fahey 2017 (sulforaphan).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).